

V3KTOR Session File Format Specification

Version 1.0 — A portable format for affective calibration session data

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License: CC BY 4.0 (this specification) · Companion document: *V3KTOR — Mapping the Self: Method, Design Rationale, and Theoretical Foundation* (Tobie David, 18 May 2026)

Abstract

This document specifies the V3KTOR session file format v1.0, a plain-text, human-readable, machine-parseable export format for affective calibration session data produced by the V3KTOR instrument. The format is designed for portability across large-language-model (LLM) interpretation contexts, for direct human inspection, and as the foundation for a future cryptographically signed personal affective profile intended for use as a personalization input to local AI agents.

The specification defines two document types: the **session export** (a single calibration session) and the **baseline export** (an aggregation over multiple sessions). Both share a common structural pattern and field semantics. The specification also defines a documented roadmap for v2 (signed, binary-encoded affective profile) which is not normative in v1 but is included to establish prior art for the longer-term format vision.

1. Scope and Status

This specification is normative for V3KTOR session and baseline exports as produced by the May 2026 reference implementation. Implementations claiming V3KTOR v1 compatibility **MUST** produce exports conforming to this specification. The specification is released under Creative Commons Attribution 4.0 International (CC BY 4.0); anyone may implement against it freely, including for commercial purposes.

The format is deliberately designed to be open, portable, and durable. Its longer-term intent is to serve as the basis for a standardized affective profile format readable by diverse AI agents on diverse devices. The free open status of this specification is itself load-bearing: the format only becomes useful as a standard if it is freely implementable by all parties.

2. Design Principles

The format design follows four principles, in order of priority:

Portability. The export is plain UTF-8 text. It can be copied to any clipboard, pasted into any chat interface, stored in any plain-text file. It does not require a parser, a player, or any specific application to read. Anyone who can read English (or French) and look at a number can read a V3KTOR export.

Human-inspectability. The user **MUST** be able to read their own data without tooling. This is a non-negotiable property: the V3KTOR thesis includes user ownership of the file, and ownership without inspectability is not ownership.

LLM-friendliness. The format is structured so that contemporary LLMs (Claude, GPT-4, Gemini, etc.) can interpret it correctly without specialized parsing. Fields are labeled, units are explicit, and the meaning of each value is documented in an accompanying handoff prefix (see §4.5).

Future-extensibility. The plain-text v1 format is forward-compatible with a planned v2 binary, signed format. v1 exports can be ingested by v2 tooling without loss; v2 outputs can be rendered back to v1

plain text for inspection. No v1 deployment becomes stranded by v2 adoption.

3. Document Types

3.1 Session export

A session export represents a single completed V3KTOR calibration session. It contains: (a) header line identifying the document type, (b) ISO date and time, (c) declared mood (if present), (d) aggregated V/A/D percentages, (e) calibrated response-time thresholds for the session, (f) word-level data partitioned into POSITIVE, NEGATIVE, CONFLICTS, and FASTEST sets, (g) pivot word and outcome if a pivot was offered, (h) footer identifying V3KTOR as the source.

3.2 Baseline export

A baseline export represents an aggregation across multiple sessions. It contains the same structural shape as a session export, with aggregated rather than per-session values, plus metadata indicating the session count, day span, and aggregation method. Baseline exports are produced from a user's saved session history and are intended for longitudinal interpretation.

4. Session Export Schema

4.1 Overall structure

A session export is a sequence of newline-separated text blocks. Each block is self-describing via a labeled header line. The canonical structure is shown below (square brackets indicate optional fields):

```
=====
V3KTOR · Session Data
=====
DATE: 2026-05-18T14:32:11
[MOOD: calm]
-----
VAD AGGREGATE: V42% · A58% · D51%
WORDS: 60
AVG RT: 743ms
RT CALIBRATION: maxRt=2000ms, fast<600ms, slow>1200ms
-----
POSITIVE WORDS (responded right, V>0.5):
  love (412ms) · peace (538ms) · home (621ms) · trust (701ms) · ...
-----
NEGATIVE WORDS (responded left, V<0.5):
  death (482ms) · alone (619ms) · betrayal (701ms) · ...
-----
CONFLICTS (response inverts word's usual charge):
  freedom(■) · grief(■) · ...
-----
FASTEST RESPONSES (under fast threshold):
  love(412ms) · death(482ms) · trust(701ms) · ...
-----
PIVOT: change → accepted
=====
V3KTOR // Mapping the Self
=====
```

4.2 Field semantics

DATE: ISO 8601 datetime of session completion (local time, no timezone offset in v1; v2 will include offset).

MOOD: optional, one of the predefined mood tokens (calm, happy, sad, anxious, angry, tired for the English version; calme, heureux, triste, anxieux, en colère, fatigué for the French version). Omitted from the export if the user skipped the mood declaration.

VAD AGGREGATE: three percentages representing the session's centroid in VAD space. V is the proportion of positive-direction responses weighted by word valence. A is the session arousal centroid weighted by per-word arousal scores. D is the session dominance centroid. Each is reported as an integer percentage 0–100. The exact aggregation formula is described in §4.3.

WORDS: count of completed responses in the session (responses with a recorded direction; aborted or timed-out words are excluded).

AVG RT: arithmetic mean of completed-response response times, in milliseconds.

RT CALIBRATION: three numeric fields documenting the response-time thresholds in effect for this session. maxRt is the per-word response window (default 2000ms; user-adjustable). fast is the threshold under which a response is classified as automatic (default 30% of maxRt). slow is the threshold above which a response is classified as friction (default 60% of maxRt). These values reflect the EWMA-blended personal calibration described in §3.1 of the companion method document.

POSITIVE WORDS: list of word-tokens for which the user responded in the positive direction, each with its response time in milliseconds. Words are separated by ' · ' (space, middle dot, space). Word order is presentation order.

NEGATIVE WORDS: as above, for the negative direction.

CONFLICTS: list of words whose response inverts the word's pre-assigned normative valence band. Each entry is followed by a direction indicator: ■ for positive direction, ■ for negative direction. A word appears in CONFLICTS in addition to its appearance in POSITIVE or NEGATIVE — the conflict set is a derived view, not a partition.

FASTEST RESPONSES: words whose response time falls below the session's fast threshold, listed with their response times. This set is a derived view; words listed here also appear in POSITIVE or NEGATIVE.

PIVOT: if a pivot word was offered during the session, this line records the pivot word and its outcome. Outcome values are 'accepted' (positive direction) or 'held at distance' (negative direction). If no pivot was offered, this line is omitted.

4.3 VAD aggregate computation

For each completed response i in the session with valence v_i , arousal a_i , dominance d_i (each on $[0,1]$) and response time rt_i :

$V = \text{round}(100 \times \text{mean}(v_i \in \text{positively-responded} \cup (1 - v_i) \in \text{negatively-responded}))$

$A = \text{round}(100 \times \text{mean}(a_i \text{ across all completed responses}))$

$D = \text{round}(100 \times \text{mean}(d_i \text{ across all completed responses}))$

These aggregations are intentionally simple: the v1 format prioritises interpretability by LLMs over statistical sophistication. v2 will add confidence intervals and weighted aggregations.

4.4 Word identifier convention

Words are emitted in their lexicon form (the surface form shown to the user). No stemming, lemmatization, or canonicalization is applied at the format layer. The lexicon is currently 368 words in the English pool across 12 thematic clusters (relational, grief, alone, transcendence, dark, light, power_dark, power_light, primal, nature, transform, pivot); the French pool is comparable in size. The

lexicon is treated as a versioned implementation detail and is not part of this format specification; readers MUST NOT depend on a specific word being present.

4.5 AI handoff prefix

When the export is produced via the 'Copy for AI' action, the session block is prepended with a structured prompt-prefix designed for LLM interpretation. The prefix is a separate concern from the format specification proper, but is described here because it is part of the V3KTOR interpretive contract.

The handoff prefix declares: (a) what the data represents; (b) the meaning of each field and unit; (c) the relationship between data and interpretation (the model is interpreting, not the instrument); (d) interpretation constraints — portrait-first opening, no metric restatement, recursion-governor for follow-ups, distress-signal close on risk patterns. The handoff prefix is normative for V3KTOR-conforming exports but is permitted to evolve in form across versions.

5. Baseline Export Schema

A baseline export aggregates across N saved sessions spanning D days. Its structure mirrors the session export with the following differences:

(a) header line identifies the document as 'V3KTOR · Baseline (aggregated data)'; (b) DATE field is replaced with DATE RANGE (earliest session timestamp — latest session timestamp); (c) SESSIONS and DAYSPAN fields document the aggregation extent; (d) VAD AGGREGATE is computed across all sessions, weighted by session word count; (e) word-level sets contain words that recurred across multiple sessions with their cumulative response counts and mean response times; (f) a STABLE SIGNATURE block lists words that the user consistently responds to in the same direction across at least 3 sessions; (g) a VARIABLE SIGNATURE block lists words whose response direction has shifted across sessions, with the proportion of each direction.

6. Forward Compatibility (v2 Roadmap)

This section is non-normative for v1. It is included to establish prior art for the format's longer-term design as of the date of this specification.

6.1 Signed affective profile

v2 of the format extends the session and baseline exports into a signed, portable affective profile. The profile is a single document (binary-encoded or JSON-encoded with canonical-form serialization) that contains: (a) a user-controlled identity key (Ed25519 or equivalent), (b) the user's baseline VAD signature across longitudinal sessions, (c) the user's stable response-time calibration, (d) the user's characteristic conflict patterns, (e) a creation timestamp and last-updated timestamp, (f) an integrity signature covering the document body, signed by the user's identity key.

The profile is designed to be self-sovereign: the user holds the signing key locally; the profile can be transmitted, stored, or shared, but cannot be modified by any party without invalidating the signature. The profile is intended to be consumed by local AI agents as a personalization input that is verifiably authored by the user and verifiably unmodified.

6.2 Local-agent consumption contract

Local AI agents consuming a V3KTOR profile SHOULD: (a) verify the integrity signature before use; (b) use the profile to inform tone, framing, and trust calibration in their interaction with the user, not to make claims about the user; (c) treat the profile as stable but not unchanging — the user may rotate or update it; (d) respect the explicit scope of the profile — V3KTOR profiles describe affective signal only, and

SHOULD NOT be treated as a general behavioural profile.

Local AI agents consuming a V3KTOR profile MUST NOT: (a) transmit the profile to remote services without explicit user consent; (b) modify the profile and re-sign it under any key other than the user's; (c) use the profile for advertising or content-recommendation purposes adjacent to its stated function.

6.3 Backwards compatibility

Any v2 implementation MUST be capable of ingesting a v1 plain-text export and producing a v2 profile from it. Any v2 implementation MUST be capable of producing a v1 plain-text rendering of a v2 profile for human inspection. This bidirectional bridge is normative for v2 compliance and is included here in v1 to lock in the compatibility commitment.

7. Authoring, Registration, and Standardization

This specification is authored by Tobie David and released under CC BY 4.0 to facilitate broad adoption. The author retains attribution rights but does not retain control rights: anyone may implement, extend, or fork the specification without permission. The author's intent is that V3KTOR become a de facto open standard for affective profile interchange, rather than a proprietary format under any single entity's control.

Future versions of this specification may be developed collaboratively. The author reserves the right to publish authoritative version-numbered editions; community forks are permitted and welcome but SHOULD use distinct version identifiers to avoid confusion.

Colophon

This document is deposited as prior art establishing the V3KTOR session file format v1.0 and the documented roadmap for v2 (signed affective profile) as of 18 May 2026. It is released under Creative Commons Attribution 4.0 International (CC BY 4.0). Implementations of this specification are unrestricted; commercial implementations are explicitly permitted.

Companion document: *V3KTOR — Mapping the Self: Method, Design Rationale, and Theoretical Foundation*, Tobie David, 18 May 2026, deposited separately. Reference implementation: *v3ktor_eng.html* and *v3ktor_fr.html*, Tobie David, 18 May 2026, released under AGPL-3.0-or-later.

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